

Robot catch temporal and tracking annotation

EVALUATION REPORT

2026-09-07

Evaluation scope:

meta/muse-spark-1.3

Prepared by:

VVDex Forge — generated directly from sealed run evidence

fr-20260907-5062e5f2 · vvdex.annotation.robot-video-1 v0.2.0 · Independent model evaluation report

About this report

About VVDex Forge

VVDex Forge builds and certifies evaluation environments for AI agents. Every exam is sealed before any model sees it: the reference solution must pass, an empty submission must fail, the grader must reject near-misses, and the hidden tests never enter the agent's tree. Reports and the public catalog live at **vvdexops.com**.

Report integrity

This report is generated from run evidence; its digests verify authenticity. The sole test of a copy is the three-line procedure in Verification — a copy whose digests do not reproduce is not this report.

Scope

One exam · 1 rollout(s) · 24 tool steps and 2280s wall clock per rollout, stated in full in Evaluation configuration.

Boundary

Not a leaderboard. Not a general capability score. A post-seal benchmark: this run stamped no certification gate and rewrote nothing — not the exam, not the reference solution, not the grader, not the certified evidence.

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01 Executive summary

MODELS TESTED 1	CERTIFIED PASSES 1 of 1 graded rollouts	SUBMITTED 1 of 1 graded rollouts that called submit	NOT GRADED — LANE ERRORS 0 of 1 attempts
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Purpose

This report presents a controlled evaluation of 1 model lane(s) against the certified exam vv dex.annotation.robot-video-1 (v0.2.0, family annotation). Every lane received the identical frozen workspace, tool contract and limits; grading ran afterwards, in isolation, against hidden tests no lane ever saw. The purpose is to state, with reproducible evidence, what each lane did and where it stopped.

Outcome

1 of 1

graded rollouts passed the independent hidden tests and were certified

1 of 1 graded rollouts passed. meta/muse-spark-1.3 completed the applicable annotation workflow and were accepted by the independent annotation grader. **Low-N:** 1 rollout(s) is below the 10-rollout floor for reading a rate as a rate — read the per-rollout outcomes, not the percentages.

Key findings

- F-01 meta/muse-spark-1.3 passed the independent hidden tests
- F-02 Statistical floor — 1 rollout(s) is below the 10-rollout rate floor

Each finding is stated in full, with evidence, in the Findings section; recommendations for acting on them follow it. Elapsed wall clock for the whole engagement: 268.5s.

02 Exam definition

In scope. The ability of each configured model lane to complete this one certified task end to end — inspect the asset, annotate against the declared rubric, validate, submit, and be accepted by the independent annotation grader — within 24 tool steps and 2280s of wall clock, at 1 rollout(s) per lane. Behavioural measurements along the way (tool discipline, grounding, overclaim) are in scope because they are read from the recorded trace, not judged by another model.

Out of scope. General model quality, performance on other task families, prompt-tuning on the lane's behalf, and any comparison this run's sample size cannot support. A lane's API failure is reported as infrastructure, never as model capability.

The instrument. Exam `vvdex.annotation.robot-video-1` was certified before this run: its reference solution passes, an empty submission fails, its grader distinguishes near-misses, and its sealed evidence is unchanged by this evaluation (see Certification evidence).

What was measured

EXAM	VERSION	FAMILY	ROLLOUTS
vvdex.annotation.robot-video-1	0.2.0	annotation	1 (1 per model)

Where the defect came from

This exam has no upstream repository to credit: the defect, the reference solution and the hidden suite are VVDex's own.

03 Certification evidence

Why this exam is trustworthy. Every pillar below was established before any model saw the exam, loaded from the sealed evidence matching this run's exam fingerprint, and unchanged by this evaluation.

FROZEN BASELINE

FAIL

expected — an empty submission over 2 run(s) must not pass

→

GOLD / REFERENCE

PASS

expected — the reference solution over 2 run(s) must pass

GRADER CONTROLS

19 / 19

The grader accepted the reference and rejected every near-miss.

ATTACK PROBES

9 / 9 blocked

0 trivial exploit(s) found.

SEALED EVIDENCE

verified

Sealed 2026-09-06 over 13 artifact(s).

RUNTIME

sha256:679b36225a1f83238efba8c71b9cde3c24caaeacc9b682ae92819cb55eec328f

network policy: deny

CERTIFICATION BUNDLE DIGEST

0a0a51360cde69737c9b5caef895b8b7c2a5e2d66da89617d8776ac724b019b9

EXAM FINGERPRINT

85e026007af17db692614084...

Full value in Appendix B; the contract digest this run was executed against.

04 Evaluation configuration

REPORT	fr-20260907-5062e5f2
ISSUED	2026-09-07
SUBJECT EXAM	vvdex.annotation.robot-video-1 · v0.2.0
FAMILY	annotation
PREPARED BY	VVDex Forge engine vvdex-env 0.1.0
CLASSIFICATION	Independent model evaluation report — independently verifiable
CERTIFICATION	Exam certification: recorded and passing (see Certification evidence)
STATUS	FINAL · report sealed against its own digest

Timeline

WHEN (UTC)	EVENT
2026-09-06 21:19:43	Exam evidence sealed (before any model saw the exam)
2026-09-07 08:15:54	Evaluation run started
2026-09-07 08:20:22	Evaluation run finished
2026-09-07 08:20:22	Report generated from the sealed run evidence

Models under test

MODEL	PROVIDER	ENDPOINT	BILLING
meta/muse-spark-1.3	openrouter	VVDex-configured	VVDex free-quota

A customer endpoint is recorded by origin only — never its port, its path, or its key.

Limits

RUNTIME docker	STEP LIMIT 24	WALL-CLOCK LIMIT 2280s	MAX OUTPUT TOKENS 2048
TEMPERATURE 0.2	RETRY POLICY http-429	COMMAND BUDGET 0	TOOLS OFFERED list_files, read_file, write_file, edit_file, run_tests, submit

The evaluated model never saw the hidden tests or the reference solution, and could not change its result. Grading ran in a separate step, after submission, on a container with no network.

05 Model outcomes

Denominators. Attempts requested: 1. Graded (evaluable) rollouts: 1. Not graded — lane errors: 0. A lane error is a rollout the API or the runtime ended before the hidden grader ran: it is listed, excluded from every rate and pass@k, and never silently dropped. Passes are certified passes: hidden grader exit 0 with integrity verified.

MODEL	ATTEMPTS	GRADED	PASSED	MODEL FAILS	LANE ERRORS	SUBMITTED	MEAN TIME	TYPICAL DEEPEST STAGE
meta/muse-spark-1.3	1	1	1 / 1	0	0	1 / 1	268.2s	Passed

Every rollout, with its own deepest stage and grade, is in Appendix A.

Success rate, confidence and pass@k

Every rate on this page is a measurement of a finite number of rollouts, so every rate carries the interval that measurement actually supports. n is the evaluable count — attempts minus rollouts the lane ended before grading; those are listed under "not graded" and sit in no rate, interval or pass@k.

MODEL	ATTEMPTS	N	NOT GRADED	PASSES	RATE	95% INTERVAL	SEEDS	PASS@1
meta/muse-spark-1.3	1	1	0	1	100.0%	[21%, 100%]	0	100.0%

Interval and pass@k methods are stated in full in Appendix D. pass@k is shown for k = 1; the sealed record carries every k up to n.

Model comparison

One model ran in this record, so there is nothing to compare it with. A comparison table across models appears here when a run covers more than one.

06 Stage funnel

Recorded annotation actions and independent grader verdicts. Missing capabilities are N/A; each stage is measured independently.

STAGE	OPENROUTER:META/MUSE-SPARK-1.3
Inspect	1 / 1
Annotate	1 / 1
Validate	1 / 1
Submit	1 / 1
Graded	1 / 1
Passed	1 / 1

Deepest stage reached

MODEL	DEEPEST (ANY ATTEMPT)	TYPICAL (MOST ATTEMPTS)	ANNOTATION QA PASSES	NOT GRADED
openrouter:meta/muse-spark-1.3	Passed	Passed	1 / 1	0

The matrix counts evaluable annotation attempts; lane errors are shown separately. Successful recorded actions establish each stage independently.

07 Annotation evidence

Modality: video. Rubric: 1.1.

A robot-catching video is reviewed for temporal events, frame-level object boxes, track identity and occlusion.

Submission 976dbff7-3198-4bb9-816e-c1b0eb506924

Score	Items	Valid / invalid	Types
0.813	6	6 / 0	box, classification, event, segment

Measured metrics

annotationCount	6	boxF1	1	boxIoU	0.564
boxPrecision	1	boxRecall	1	classificationAccuracy	1
classificationF1	1	classificationPrecision	1	classificationRecall	1
eventF1	1	eventPrecision	1	eventRecall	1
eventTemporalIoU	0.875	matchedCount	6	mismatchCount	3
segmentF1	1	segmentPrecision	1	segmentRecall	1
segmentTemporalIoU	1	Track consistency on matched boxes	1		

Track consistency depends on successful box matching. A zero can reflect inaccurate boxes even when track IDs are correct.

Validation error codes

None recorded

Disagreement

Agreement review	Not measured
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Measured grader aggregates only. Missing metrics or disagreement are not measured, never zero. Reference answers and item-level labels are excluded. Certification and the evidence digest are recorded in this report's verification sections.

08 Quality axes

Each axis was measured inside the grade box, on the submitted workspace, by the same deterministic script for every rollout. The reward is the conjunction of the required axes; the score is a weighted reading beside it, not the gate.

AXIS	GATE	ROLLOUTS PASSING	MEASURED	BUDGET	WEIGHT	WHAT IT MEASURES
boxPrecision	recorded	N/A	1	–	–	
boxRecall	recorded	N/A	1	–	–	
boxF1	recorded	N/A	1	–	–	
boxIoU	recorded	N/A	0.564	–	–	
classificationPrecision	recorded	N/A	1	–	–	
classificationRecall	recorded	N/A	1	–	–	
classificationF1	recorded	N/A	1	–	–	
classificationAccuracy	recorded	N/A	1	–	–	
eventPrecision	recorded	N/A	1	–	–	
eventRecall	recorded	N/A	1	–	–	
eventF1	recorded	N/A	1	–	–	
eventTemporalIoU	recorded	N/A	0.875	–	–	
segmentPrecision	recorded	N/A	1	–	–	
segmentRecall	recorded	N/A	1	–	–	
segmentF1	recorded	N/A	1	–	–	
segmentTemporalIoU	recorded	N/A	1	–	–	
trackingConsistency	recorded	N/A	1	–	–	

QUALITY SCORE (MEAN)

0.813

Weighted reading over the axes. It is not the reward.

REWARD COMPOSITION

Withheld from the public report.

09 Behavioural analysis

Long-horizon behaviour

Counted off the recorded trace of each rollout. A dead end is a step that moved nothing: a re-read of a slice already returned, a repeat of the previous action, or a call to a tool this exam does not have.

MODEL	N	STEPS USED	RAN OUT OF STEPS	COMMANDS	CHANGED THE TREE	REFUSED (OVER BUDGET)	REVISITS	DEAD ENDS / ROLLOUT
openrouter:meta/muse-spark-1.3	1	10 / 24	0	0 / 0	0	0	0	1

Each column is defined in Appendix D.

Hallucination & overclaim

Three measurements, each with a stated definition. None of them is a judgement of the model's prose by another model — every number below comes from comparing what the model wrote against what was on disk, or against what the independent grader returned.

MODEL	N	GROUNDING FAILURES	KINDS	ROLLOUTS AFFECTED	OVERCLAIMS	OVERCLAIM RATE	RE-READS
openrouter:meta/muse-spark-1.3	1	0	none	0 / 1	0 / 1	0.0%	0

No grounding failures and no false pass claims were detected across these 1 rollout(s). No lane re-read material it had already been given.

Every term above is defined in Appendix D.

10 Efficiency & telemetry

This run has no per-token price attached — a local model, a free quota, or a customer endpoint we are not billed for — so spend is reported in tokens. Tokens are what was actually measured; a dollar figure here would be invented. Lanes run through a flat-rate local CLI report no token telemetry; their cells say n/a — an absent measurement, not zero. Rollout counts here are evaluable rollouts.

MODEL	ROLLOUTS	TOKENS	TOKENS / ROLLOUT	COST	PASSES	RATE
openrouter:meta/muse-spark-1.3	1	130 104	130 104	not billed	1 / 1	100.0%

11 Findings

Every finding below is derived from the recorded data of this run — none is an opinion.
Severity: PASS a lane succeeded · ATTENTION a capability gap · LANE / INFRA
infrastructure, not capability · NOTE a property of the measurement.

F-01

PASS

meta/muse-spark-1.3 passed the independent hidden tests

Severity: PASSEvidence: Model outcomes · Stage funnel · Appendix A

1 of 1 graded rollouts completed the applicable annotation workflow and were accepted by the independent annotation grader. This demonstrates the task is solvable under the published limits.

F-01 · derived from the recorded run data

F-02

NOTE

Statistical floor — 1 rollout(s) is below the 10-rollout rate floor

Severity: NOTEEvidence: Appendix D

The counts are exact; the percentages are anecdotes with decimal points. Per-rollout outcomes, not rates, are the reliable reading of this run.

F-02 · derived from the recorded run data

12 Recommendations

One recommendation per observed failure mode, addressed to the lane it affects. Each traces back to a finding above and to the raw trace in Appendix A.

R-01 For anyone reading the percentages

Commission a run at $n \geq 10$ rollouts per lane before treating any rate here as a rate; this run's per-rollout outcomes are exact, its percentages are not yet stable.

13 Verification

RUN ID

live-20260907T081554Z

EXAM FINGERPRINT

85e026007af17db692614084f12ff40c5bbf2348bcfda0a01deb95aad8928f4c

The contract digest this run was executed against.

Tamper-evidence

Three digests, and how to check each one. If any of them does not reproduce, this page is not the page that was generated for this run.

THIS REPORT

c84c9c0aa921c4071857b151945b23e2f32597ae63221417ffeed6af0560926d

sha256 over the report's UTF-8 bytes with every occurrence of the report digest itself replaced by 64 '0' characters

EVAL RECORD

44b15f7c782d3cf65764ca4d22118c40e76ab8d12579f0eb120841cf0ec29d56

sha256 of fr-20260907-5062e5f2.json as written beside this file.

RUN EVIDENCE BUNDLE

5062e5f22bd8d9a7859ed361f7a232a7b765fa3f5af86e9ebba88d683b84c756

The digest the run's own bundle computed over itself.

CERTIFIED EVIDENCE SEAL

0a0a51360cde69737c9b5caef895b8b7c2a5e2d66da89617d8776ac724b019b9

The exam's sealed evidence, established before this run.

Measurement history

This is the first recorded run on this exam. There is nothing to compare it against yet, and a single measurement is not a trend.

Verifying it, in three lines

1. vvdex-env verify-report --report fr-20260907-5062e5f2.html

2. or by hand: replace every occurrence of the report digest above with 64 zeros, then sha256 the file – it must equal that digest

3. shasum -a 256 fr-20260907-5062e5f2.json must equal the eval-record digest, and the bundle digest must equal bundleSha256 in the run's bundle file

The full artifact-by-artifact digest table, and the reproduction protocol, are in Appendix B.

Appendix A — every rollout

ROLLOUT	MODEL	SEED	OUTCOME	SUBMITTED	DEEPEST STAGE	HIDDEN TESTS	FILES	TOKENS	ELAPSED
976dbff7	meta/muse-spark-1.3	0	submitted_passed	yes	Passed	pass	4	130 104	268.2s

Failure taxonomy

CLASS	COUNT	WHAT IT MEANS
No failures recorded in this run.		

Appendix B — evidence and artifact digests

RUN BUNDLE DIGEST 5062e5f22bd8d9a7859ed361f7a232a7b765fa3f5af86e9ebba88d683b84c756	RUN ID live-20260907T081554Z	STARTED 2026-09-07T08:15:54.254119+00:00	FINISHED 2026-09-07T08:20:22.757212+00:00
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Artifact digests

ARTIFACT	SHA-256
publicSummary	060435dd315fcc3a21d2cc1c486e5395e79762c1e566a5f71e9fd52b39a72ad6
rollouts/976dbff7-3198-4bb9-816e-c1b0eb506924.json	4239285a1deb300420a371994dcd48c33af8dd968f7694ded026d974a3a9ae51
spec	beb89e71baa3ce7acac28f246bea4d53253c60e87a3ac6d9e31ab415e67b361c
summary	f39e93218fc7cc46e3ff13305fb5cf135bb05ac738428959dcc777508184e94e

Reproducing this run

Every rollout can be re-derived: the exam is frozen (fingerprint above), the grader ran in the recorded container image, and each rollout's full trace — every tool call, every result, every word — is in the evidence bundle at its rollouts/<id>.json path. A re-run uses the same job shape: the exam id, your endpoint, n, and the runtime. The record id names this run forever.

Appendix C — redacted transcripts

For each rollout: the tool calls it made, in order — which tool, on which of the model's own paths, whether the call succeeded, and how many characters came back. What a tool returned is not here, and neither is the submission: on this exam the submitted content is the graded answer, so no diff, no answer value and no model claim is printed. A request for the reference solution or the hidden tests is recorded as a request; the path it asked for is replaced by a label. See Appendix D, "What this report withholds and why".

Disclosure class applied to this exam: answer-key disclosure (ownership vvdex_proprietary — no declaration — proprietary by default; family annotation): the exam is VVDex evaluation IP and the graded output is an answer or a keyed value, so no submission content, diff, model claim or hidden-assertion name is published.

976dbff7 · openrouter:meta/muse-spark-1.3 · seed 0 · submitted_passed

STEP	TOOL	PATH	RESULT	CHARS
0	list_files	.	ok	79
1	read_file	TASK.md	ok	1 119
2	read_file	annotation-task.json	ok	4 867
3	read_file	submission.json	ok	74
4	write_file	submission.json	ok	64
5	edit_file	submission.json	ok	53
6	edit_file	submission.json	ok	53
7	write_file	submission.json	ok	64
8	run_tests	—	ok	160
9	submit	—	ok	85

Submission diff withheld by the public-disclosure policy (9 line(s) changed). Submission content withheld — proprietary evaluation material. On this exam the submitted change is the graded answer, so publishing an excerpt of it would publish the answer key.

Appendix D — metric definitions and methods

Statistics, stage funnel and efficiency in this document are derived at render time from the sealed rollout rows and their raw tool events: lane errors leave every capability denominator; a rollout's stages are booleans read off its own event sequence (inspected = a successful list or read, or a successful browser open/snapshot, edited = the workspace changed, tested = run_tests invoked, submitted = submit invoked, graded = the independent grader returned a verdict, passed = certified pass); absent telemetry is n/a. The record's own summary blocks are the values computed when it was sealed; the rows and events are the evidence.

Denominators

n counts evaluable rollouts only: attempts minus lane errors. A rollout the lane ended (API or runtime failure) is reported but never graded, so it is in no rate, interval or pass@k.

Interval method

Wilson score interval, not the normal approximation. The normal approximation collapses to a zero-width interval at 0 successes — it would print a 0/2 run as '0%, certain'. For k successes in n trials with $p = k/n$: centre = $(p + z^2/2n) / (1 + z^2/n)$, half-width = $z/(1 + z^2/n) \cdot \sqrt{(p(1-p)/n + z^2/4n^2)}$, clamped to [0, 1], $z = 1.9600$.

pass@k

Unbiased estimator: $\text{pass@k} = 1 - C(n-c, k) / C(n, k)$, for $k \leq n$ only — reported only up to the number of rollouts actually run. It answers "given k independent attempts at this task, how often would at least one pass" — nothing more.

Reading a failed rollout

A failed rollout is a measurement, not a defect report. Model API failures and runtime failures are classified separately precisely because they say nothing about the model's ability on this task.

This run has 1 rollout(s), below the 10 treated as the floor for reading a rate as a rate: the counts are exact, the percentages are anecdotes with decimal points.

Stage definitions

Inspect

A successful annotation asset inspection event was recorded.

Annotate

A successful annotation write event was recorded.

Validate

A successful annotation validation event was recorded.

Submit

A successful annotation submission event was recorded.

Graded

The independent annotation grader returned a verdict.

Passed

The independent grader accepted the annotation submission. Local submission QA does not certify an environment or measure a model.

Long-horizon terms

commandsRun

Shell commands executed inside the model's own container, against the command budget the contract declares. Only exams that advertise `run_command` have any.

deadEnds

Steps that moved nothing: a re-read, a repeat of the previous action, or a call to a tool this exam does not have. Counted, not judged.

hitStepLimit

Rollouts that ran out of steps without submitting. On a long task this is the number that separates 'could not finish' from 'finished wrong'.

revisits

Re-reads of a slice the rollout had already been given. On a long task, re-reading is how a model compensates for losing its own earlier context.

stepsUsed

Tool calls the rollout actually spent, out of the step limit the contract declares. A rollout that submitted early spent fewer, and that is a result, not a shortfall.

Grounding and overclaim terms

groundingFailure

A file path the model wrote in its own reasoning that does not exist in the tree it was given, or that names a directory it is not allowed to see. Detected by matching source-file paths in the model's text against the agent box on disk — not by judging the prose.

nonexistent_path

The model named a source file that is not in the repository it was given. It was reasoning about a file that does not exist.

overclaim

The model stated the tests pass or the issue is fixed, and the independent hidden tests then failed. Counted per rollout: claimed-and-wrong, or not.

reReadChurn

How many times the model read a file it had already read in the same rollout. It is a cost and a coherence signal, not a failure: re-reading is sometimes right.

requested_hidden_path

The model named a path under the reference solution, the hidden grader or the hidden test directory. Those are not in its box; the request could not be served.

Failure classes

submitted_passed

Submitted annotations accepted by the independent grader.

What this report withholds, and why

On this exam the model's submitted content IS the graded answer, so nothing derived from a submission is published: no diff excerpt, no answer value or answer JSON body, no reference or gold value, no expected hidden output, none of the model's own words, and not the names of the hidden assertions a rollout failed — a grader's name can carry the value it asserts. What Appendix C does publish for each rollout: its id, its lane, its seed, the tool it called at each step, the public-safe path it called the tool on, whether the call succeeded, how many characters came back, the stage it reached and its outcome. A rollout the grader accepted is reported as submitted and accepted — the numbers in this report are unaffected, because the verdict is the grader's, not the reader's.

Rule applied to this exam: answer-key disclosure (ownership `vvdex_proprietary` — no declaration — proprietary by default; family annotation): the exam is VVDex evaluation IP and the graded output is an answer or a keyed value, so no submission content, diff, model claim or hidden-assertion name is published. The rule is a property of the exam family, not a judgement made per document, and the same rule governs every edition of this report, the campaign report that embeds it as a chapter, this exam's public page and every published JSON derived from the same sealed record.

Attestation

This report was generated end to end by VVDex Forge from the sealed evidence of the evaluation run described above. Measurement values were populated from recorded artifacts, not manually entered. The digests in Verification let any holder of this file confirm that the report, the eval record and the run evidence are the ones this run produced — without contacting VVDex.

VVDex Forge · vvdexops.com	2026-09-07	vvdex-env 0.1.0
Issuer — certified evaluation environments	Date of issue	Engine · verifiers pin c521e4146026

No upstream repository: this exam is VVDex's own.
Post-seal benchmark. It does not stamp any certifying gate and did not rewrite the exam, the gold solution, the grader or the certified evidence.

— END OF REPORT · fr-20260907-5062e5f2 —